自己紹介

ナイルファーズ

Nailfaaz

ナイル 概要



プロフィール

名前：ナイルファーズ
大学：ガジャマダ大学
　　コンピュータサイエンス
卒業：2026 年 3 月（優等）
GPA：3.87 / 4.0
現在：AI エンジニア
言語：インドネシア語・英語

技能

AI / ML：PyTorch, scikit-learn, OpenCV, YOLO
LLM：LangChain, RAG, Qdrant, Langfuse
データ：SQL, Pandas, BigQuery, 時系列予測
インフラ：Docker, Kubernetes, Azure, FastAPI

活動歴

仕事

- **AIエンジニア** — Kalbe Farma・2026.02〜
- **データサイエンス** — GoTo・2024.09〜25.03
- **AI五輪メンター** — hAIppy.id・2026.01〜

受賞歴

- **第1位・ITB** — データサイエンス・2026.05
- **第1位・UMN** — データサイエンス・2024.12
- **第1位・UI** — データソン・2023.08
- **第2位・UNY** — データマイニング・2024.05
- **全国大会本選** — GEMASTIK・2024.09
- **全額奨学金** — Tanoto Foundation

論文

- **ICTIS 2026** — Springer・胸部X線分割
- **ICST 2024** — Trans Tech・違法駐車
- **ICIC 2023** — IEEE・都市緑地の分割

学生団体

- **Kotak Riset（AI研究室）** — 研究メンバー → 代表・2023.04〜25.10
- **OmahTI（データ・AI部）** — メンバー → 副部長・2022.12〜24.12

学び続ける力
新しい技術を
すぐに試して吸収

円滑な連携
立場の違う人と
丁寧に調整

両立力
複数の仕事を
計画的に進める

興味

- 研究と実装の橋渡し
- LLM・生成 AI
- コンピュータビジョン
- データ分析

趣味

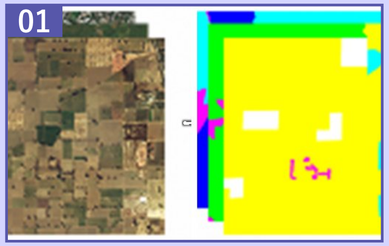
- さんぽ
- ジャーナリング
- ゲーム

目標

- 学び続ける
- 日本で長期的にキャリアを築く
- 現場で使われるシステムを作る
- チームを率いるエンジニアに

プロジェクト 一覧

01



衛星画像の土地被覆分類

Satellite land-cover segmentation, 7 classes, 83.9% pixel accuracy

PyTorch / DeepLabV3+

p.4

02



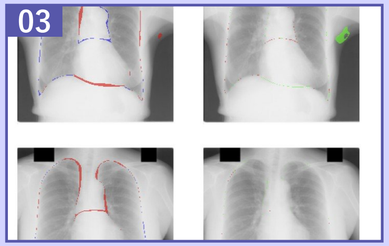
医療保険クレームの予測

Frequency × severity forecasting; 1st place, MCF ITB 2026

時系列予測 / NumPy

p.6

03



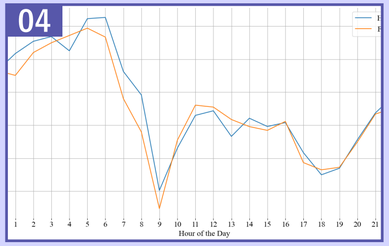
胸部 X 線の多臓器分割

Thesis: U-Net vs DeepLabV3+, CRF-as-RNN; ICTIS 2026 paper

PyTorch / 論文

p.8

04



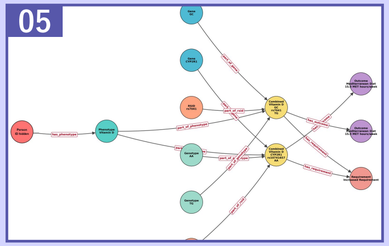
交通速度の予測

934 London roads, one model per road; Random Forest best

scikit-learn / Datathon

p.10

05



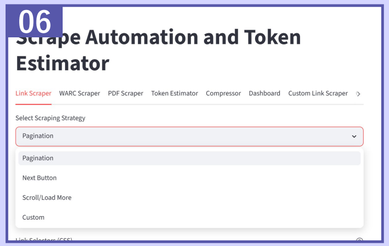
RAG チャットボット基盤

Ingestion, memory and tracing for production chatbots

LLM / Azure / K8s

p.12

06



Ez-Scrape データ収集

Web and PDF scraping for LLM training data (600K+ files)

Selenium / Streamlit

p.14

07



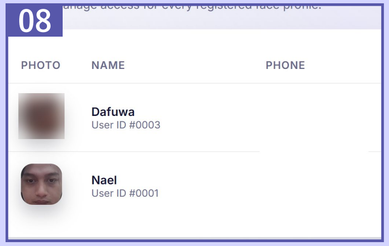
違法駐車検知システム

YOLOv8 + ByteTrack, ≈20 FPS; 1st place, Datathon 2023

CV / YOLOv8

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08



顔認証入退室管理システム

Face login with InsightFace and an admin dashboard

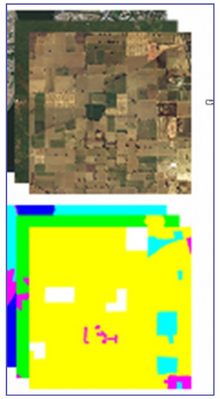
Flask / InsightFace

p.18

衛星画像による土地被覆セグメンテーション

Kaggle › nail2004/land-cover-segmentation

目的	:	Investigated stratified data splitting to handle class imbalance in multi-class segmentation of satellite imagery (land-cover classification).
言語	:	Python
技術	:	PyTorch, Python Imaging Library (PIL), Kaggle
メンバー｜期間	:	3 Person｜3 weeks
自分の役割	:	Data preprocessing and data splitting (stratification)
アウトプット	:	Land-cover segmentation model for 7 classes with 83.9% pixel accuracy (DeepLabV3+, stratified split, TTA + multi-scale inference).

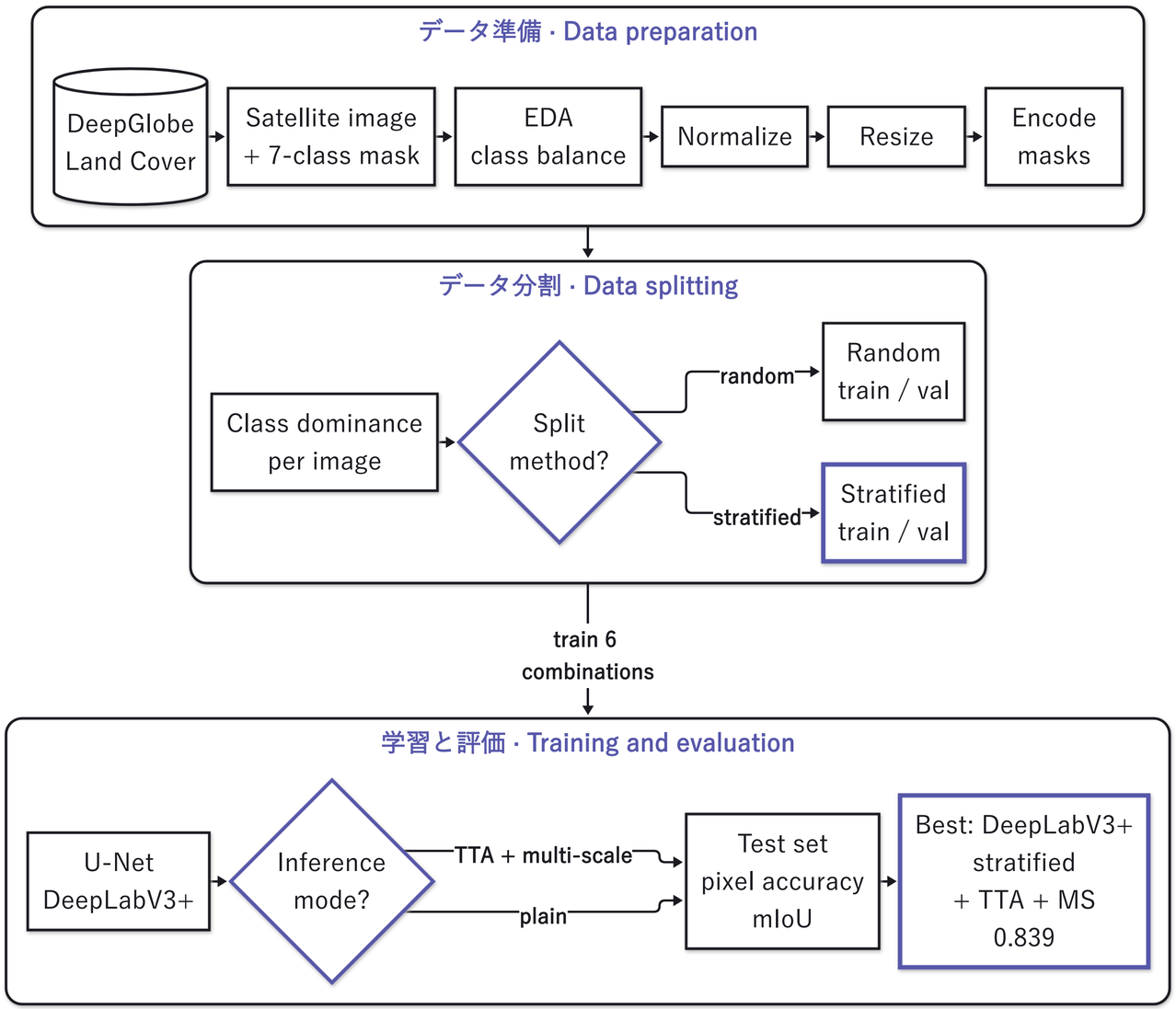


Satellite image and its annotated mask

Architecture	Split	Enhanced	Pixel acc.	mIoU
U-Net	random	–	0.758	0.584
U-Net	stratified	–	0.728	0.545
U-Net	stratified	TTA + MS	0.775	0.584
DeepLabV3+	random	–	0.802	0.617
DeepLabV3+	stratified	–	0.818	0.627
DeepLabV3+	**stratified**	**TTA + MS**	**0.839**	**0.641**

Test set. TTA = test-time augmentation, MS = multi-scale inference.

衛星画像による土地被覆セグメンテーション



データ Dataset

DeepGlobe Land Cover: RGB satellite images (2448 × 2448 px, 50 cm per pixel) labelled with 7 classes: urban, agriculture, rangeland, forest, water, barren and unknown.

結果 Findings

- DeepLabV3+ beats U-Net in every setting.
- Stratified split: +1.6 pts pixel accuracy for DeepLabV3+, but −3.0 pts for U-Net.
- TTA + multi-scale adds +4.7 pts (U-Net) and +2.1 pts (DeepLabV3+).

応用 Relevance

Land-cover maps from satellite imagery are a typical input for screening land, e.g. for solar development.

医療保険クレームの予測（頻度 × 平均請求額）

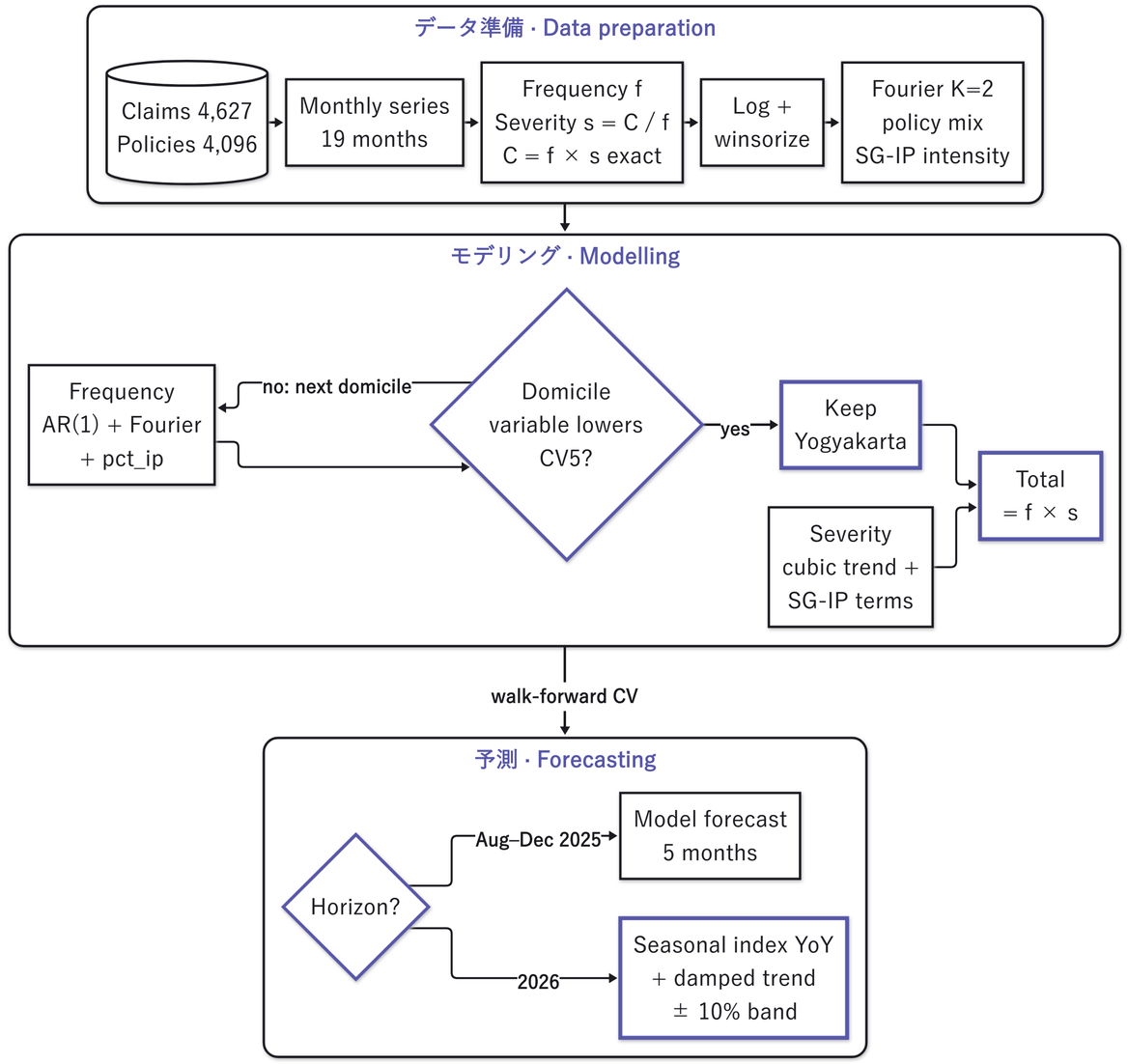
目的	:	Forecast monthly health-insurance claims (frequency, severity and total) for AXA Financial Indonesia.
言語	:	Python
技術	:	pandas, NumPy, scikit-learn, Matplotlib; ARIMA and regression models written in NumPy
メンバー｜期間	:	3 Person｜2026.02 – 2026.03
自分の役割	:	Frequency–severity modelling and forecasting
アウトプット	:	1st place, MCF ITB 2026 Data Science Challenge. Forecasts to Dec 2026 with a ±10% band.





Total claim per month (Rp billion): history to Jul 2025, forecast Aug–Dec 2025 and 2026.

医療保険クレームの予測（頻度 × 平均請求額）



検証 Validation

Fold	Months	MAPE
CV1	Jul 2025	9.31%
CV2	Jun–Jul	7.45%
CV3	May–Jul	14.06%
CV4	Apr–Jul	18.83%
CV5	Mar–Jul	15.44%

Walk-forward, one month ahead: mean of the frequency, severity and total MAPE.

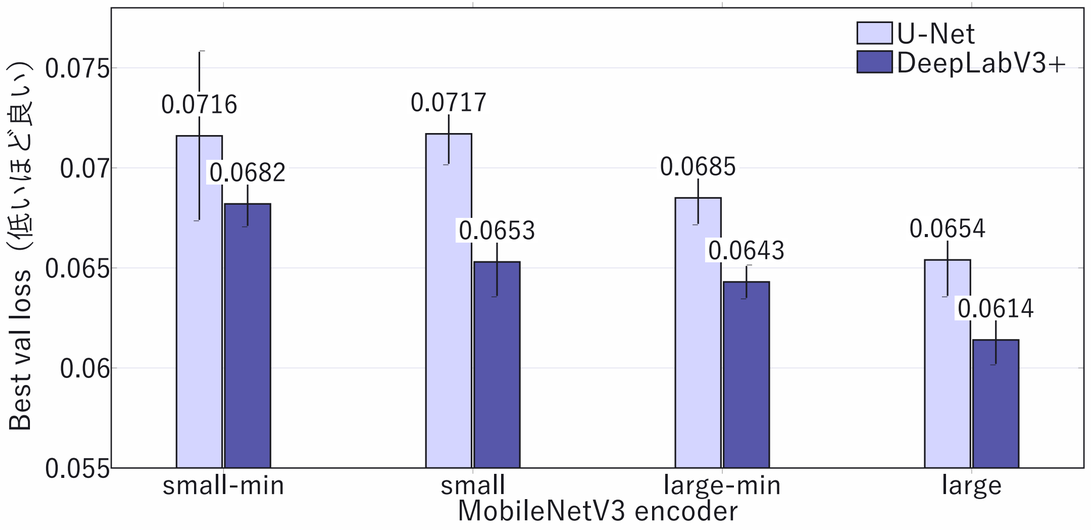
発見 Findings

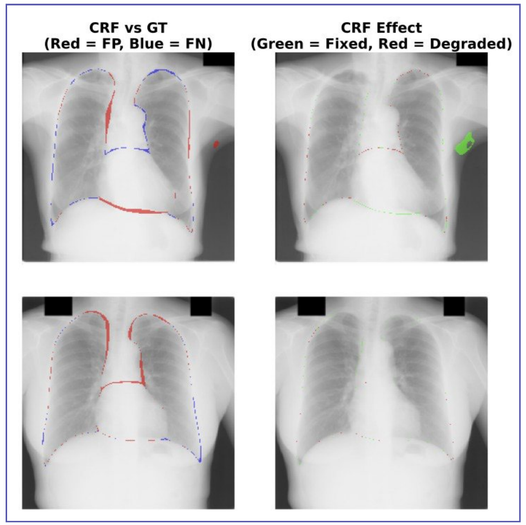
- Top 5% of claims carry 45% of the value. Singapore inpatient claims average Rp 280M each.
- Severity explains 65.6% of the variance in total claims, so it got the most modelling effort.
- 19 months of history for a 17-month horizon: forecasts are damped and banded.

胸部 X 線の多臓器セグメンテーション（卒業研究）

GitHub › Nailfaaz/skripsi

目的	:	Segment the left lung, right lung and heart in chest X-rays, and find which architecture and encoder size works best before adding CRF-as-RNN refinement.
言語	:	Python
技術	:	PyTorch, segmentation_models_pytorch, OpenCV, Kaggle GPU
メンバー｜期間	:	1 Person (undergraduate thesis)	2025 – 2026
自分の役割	:	Experiment design, training pipeline, evaluation, paper writing
アウトプット	:	Paper accepted at ICTIS 2026 (Springer LNNS). Baseline: DeepLabV3+ with MobileNetV3-large is best (val loss 0.0614 ± 0.0012).





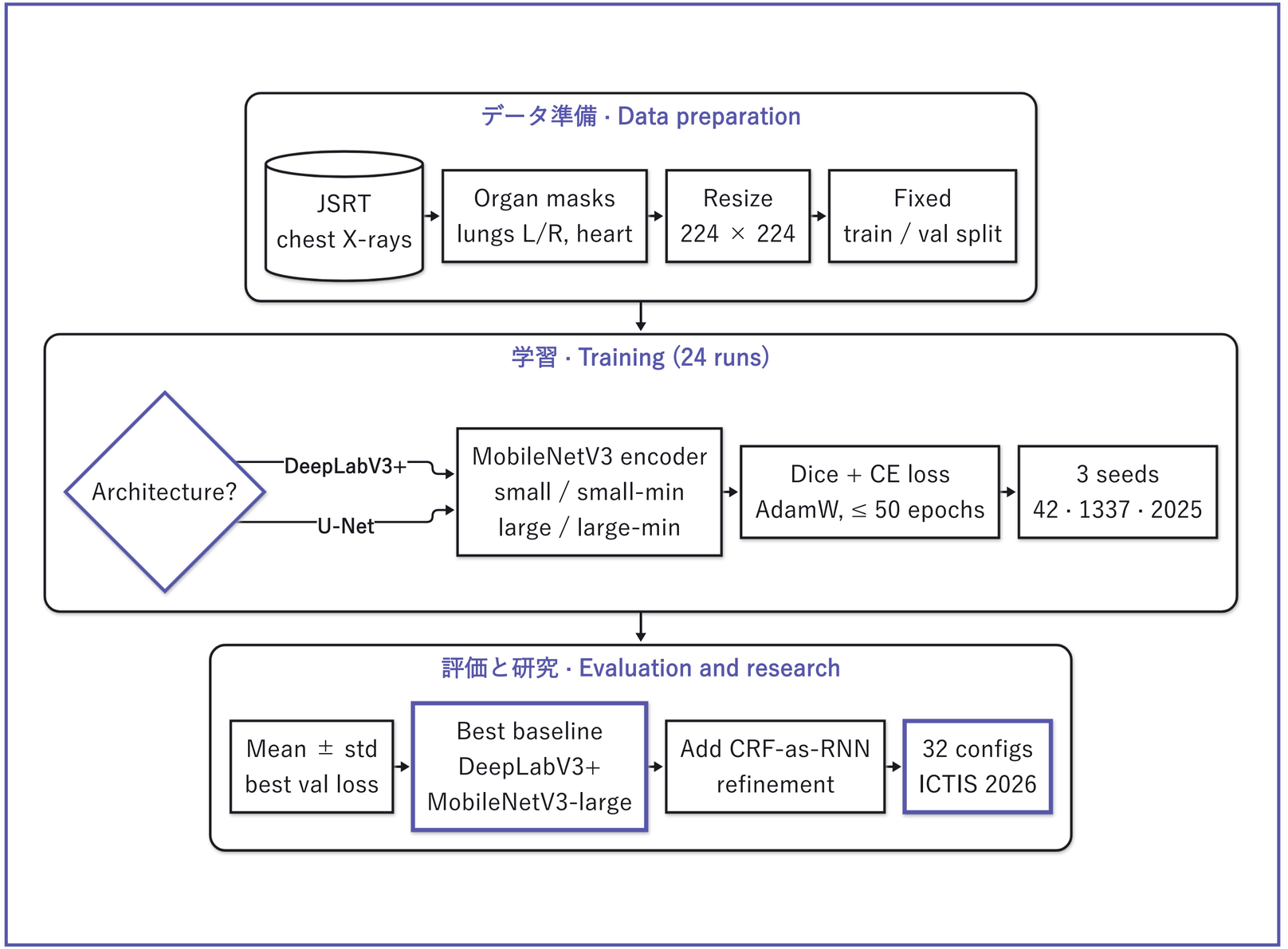
論文 Paper

Systematic Evaluation of CRF-as-RNN for Multi-Organ Chest X-Ray Segmentation

ICT for Intelligent Systems: Proceedings of ICTIS 2026, LNNS, Springer Nature (accepted, 2026.01)

← CRF-as-RNN on two test X-rays: errors vs. ground truth (left) and where CRF fixed or degraded the mask (right).

胸部 X 線の多臓器セグメンテーション（卒業研究）



設定 Setup

- JSRT chest X-rays, 4 classes (background, left lung, right lung, heart), 224 × 224.
- U-Net and DeepLabV3+ × 4 MobileNetV3 encoders × 3 seeds = 24 runs, about 1.5 min each.

結果 Findings

- DeepLabV3+ beats U-Net with every encoder.
- Larger encoders help both architectures.
- Paper: across 32 configurations, CRF-as-RNN gains shrink once the network is large enough.

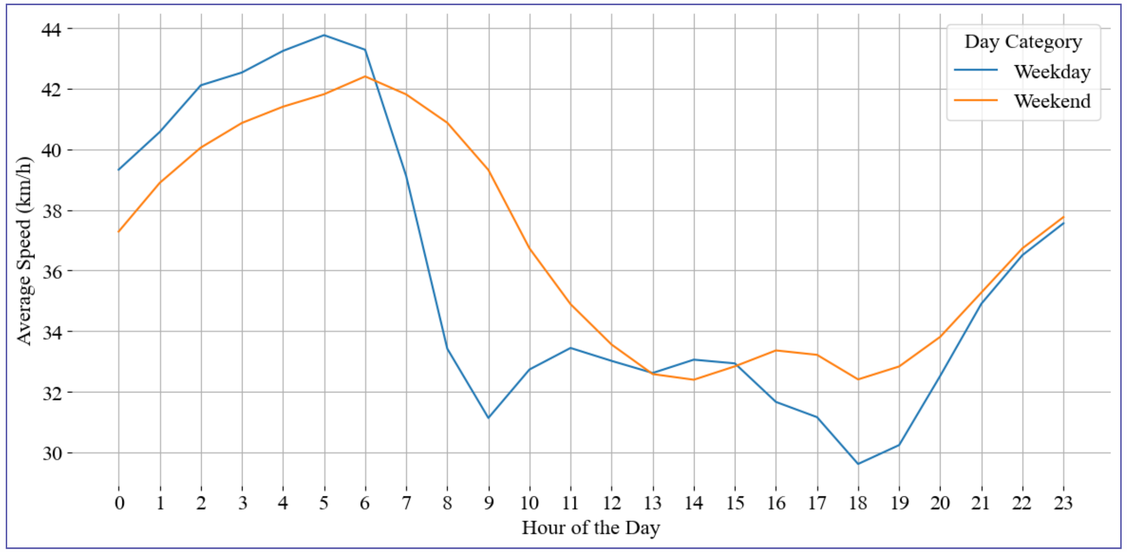
次の段階 Next

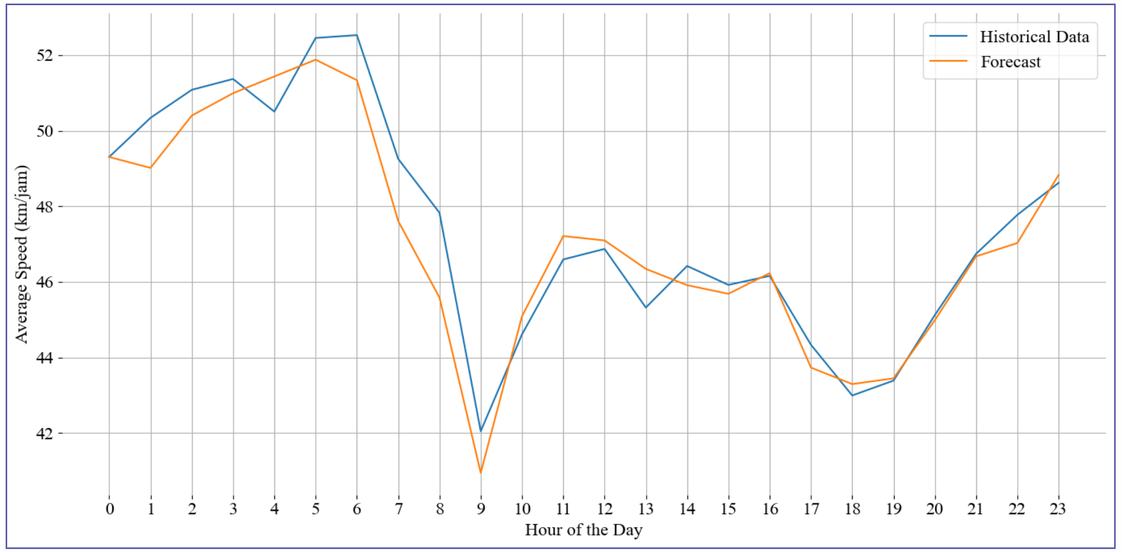
- Per-organ Dice and IoU on the held-out test set.
- Upload the CRF-as-RNN experiments to the repository.

交通速度の予測（Waze データ）

GitHub › Nailfaaz/Future-Traffic-Speeds

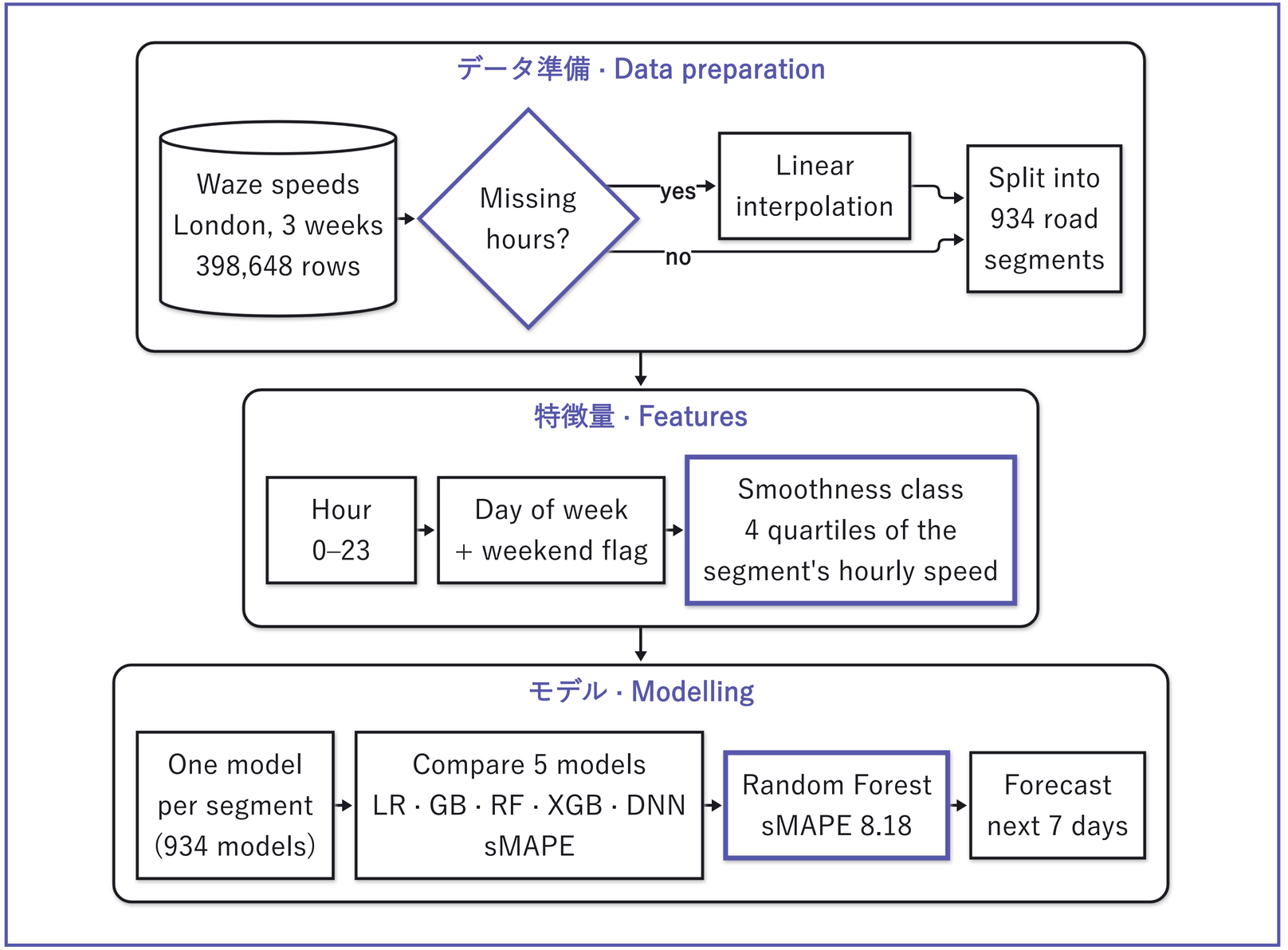
目的	:	Forecast the hourly average speed on 934 London road segments for the next 7 days from 3 weeks of Waze data.
言語	:	Python
技術	:	pandas, NumPy, scikit-learn, XGBoost, Keras
メンバー｜期間	:	3 Person｜RISTEK Datathon 2023, Universitas Indonesia
自分の役割	:	EDA, feature engineering, modelling
アウトプット	:	One model per road segment; Random Forest best of 5 models (sMAPE 8.18).





Left: average speed by hour, weekday vs weekend.　　Right: forecast vs actual for one segment.

交通速度の予測（Waze データ）



モデル比較 Models

Model	sMAPE
Random Forest	**8.18**
XGBoost	8.23
Gradient Boosting	8.58
Deep Neural Network	8.71
Linear Regression	9.45

発見 Findings

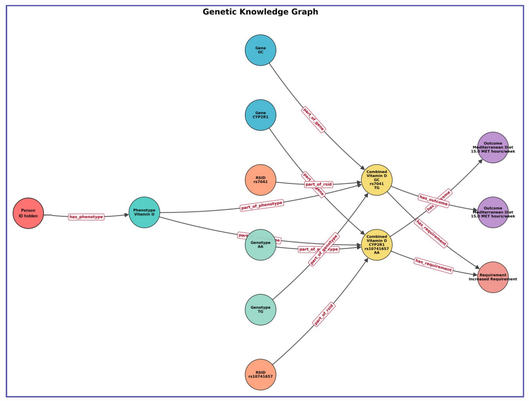
- Rush hours at 06:00–09:00 and 17:00–19:00; the morning dip is much sharper on weekdays.
- The smoothness class is a top-5 feature in all 934 models (mean importance 0.60).

工夫 Design choice

- One model per road: each learns its own traffic pattern, and a road can be retrained alone when its traffic changes.

RAG チャットボット基盤の開発

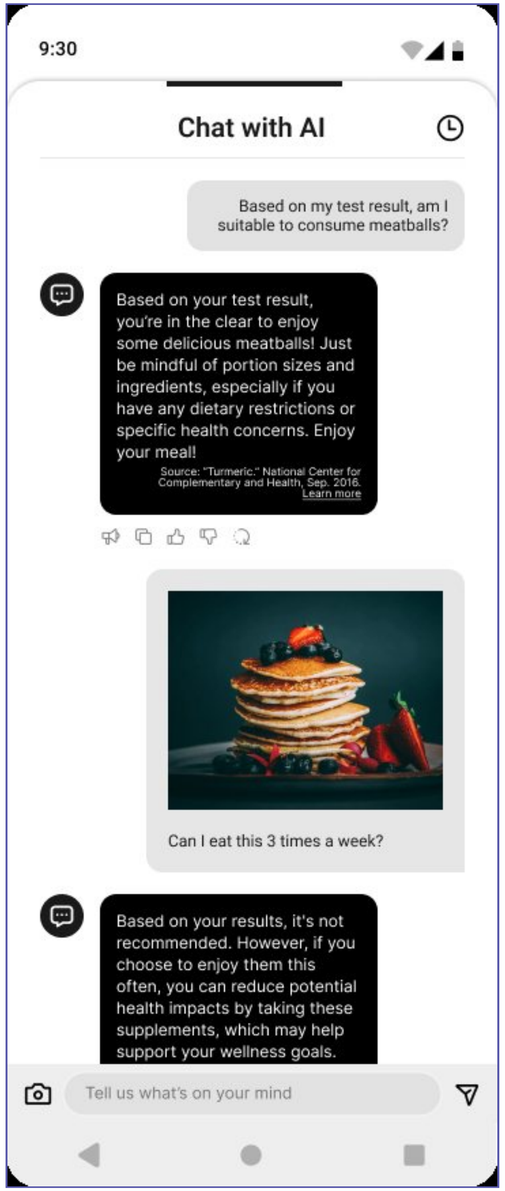
目的	:	Build the ingestion, memory and observability layers behind a production RAG chatbot for health questions.
言語	:	Python
技術	:	FastAPI, Celery, ArangoDB, Qdrant, Langfuse, Arize Phoenix, Docker, Azure AKS, OpenShift
メンバー｜期間	:	AI Engineer Intern, Corporate Digital Technology ｜ 2026.02 – present
自分の役割	:	Ingestion, memory, observability and Kubernetes deployment
アウトプット	:	Faster ingestion and memory writes, end-to-end tracing, manifests validated on a live cluster.



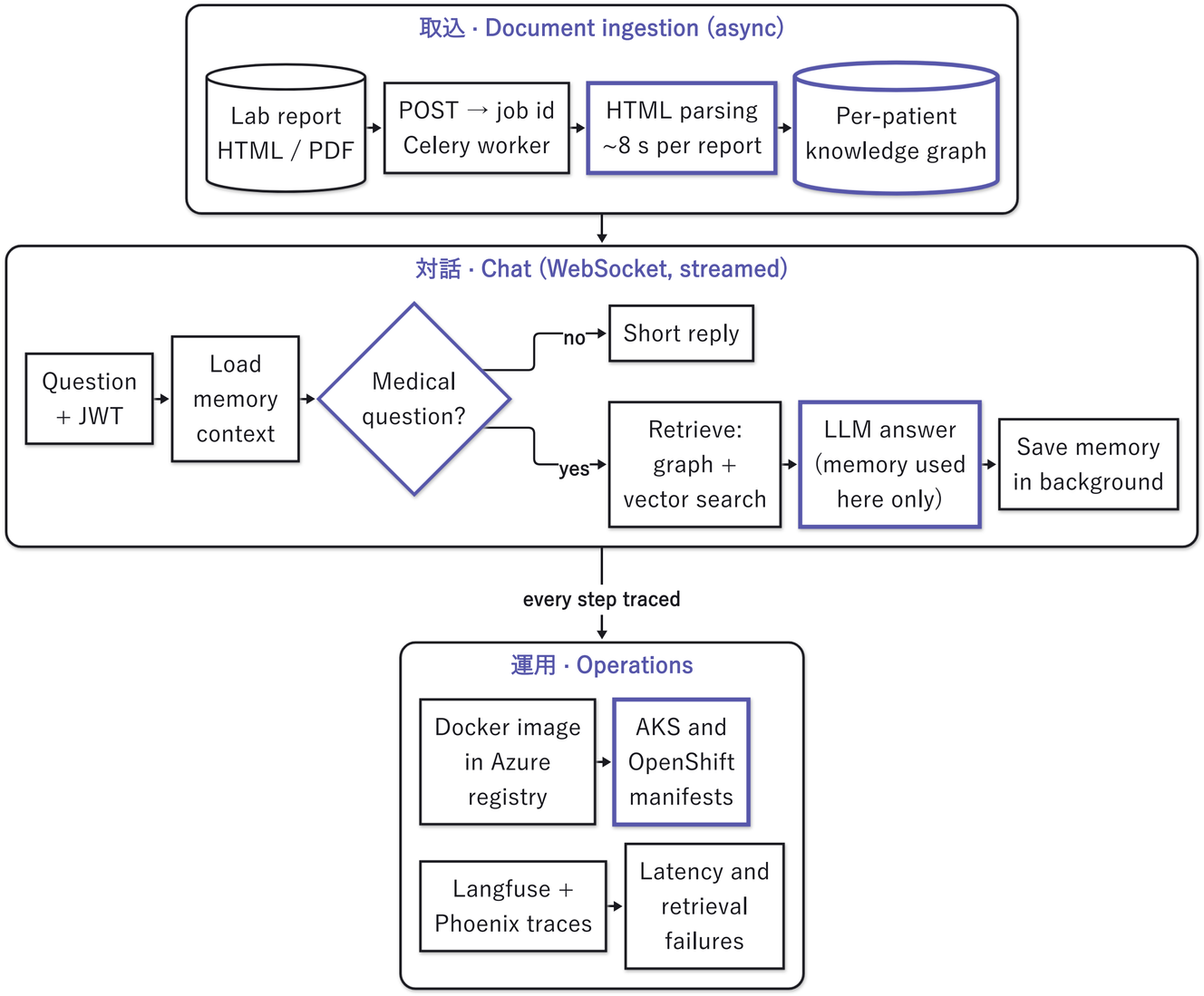
Per-patient knowledge graph built by the ingestion pipeline.

仕組み How it works

- Lab reports become a graph per patient: phenotype, gene, genotype, outcome and requirement.
- The chatbot answers from that graph and a nutrition database, and remembers earlier turns.
- Users can attach a photo of a meal; each answer cites its sources.



RAG チャットボット基盤の開発



記憶 Memory

- Short-term facts in a vector store, one collection per user.
- Long-term knowledge-graph triples in a graph database.
- Memory is injected only at the answer step and saved in the background, so streaming is never blocked.

取込 Ingestion

- Asynchronous: the API returns a job id at once; a Celery worker does the extraction and clients poll the status.

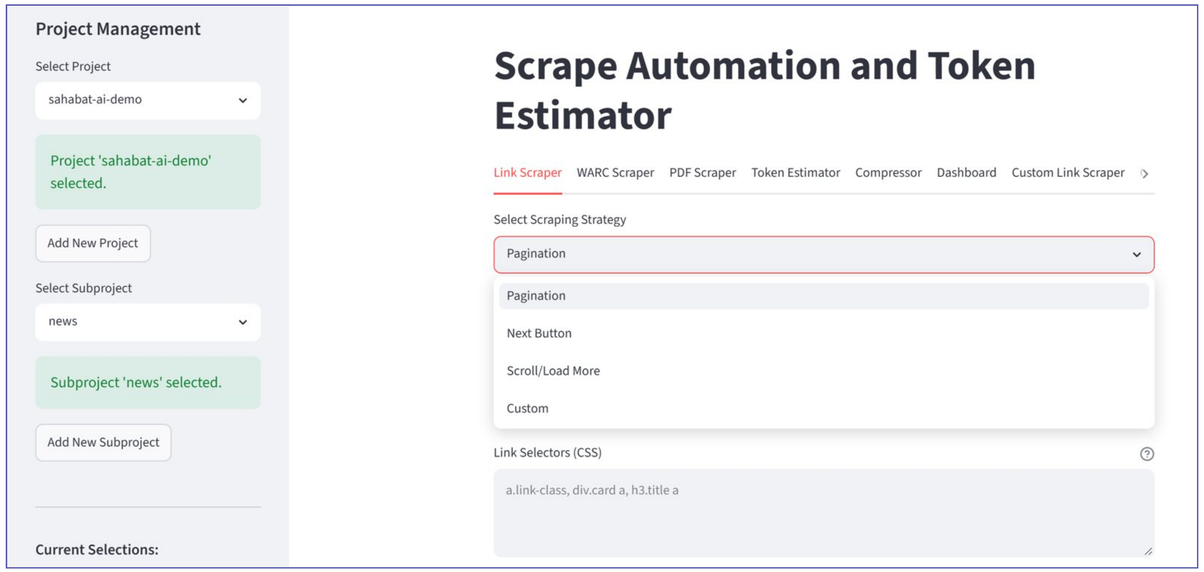
配信 Delivery

- Answers stream over WebSocket with status updates every 2.5 s; token-based login.
- Two services are deployed: ingestion, and chatbot with memory.

Ez-Scrape：LLM 学習データ収集アプリ

GitHub › Nailfaaz/ez-scrape

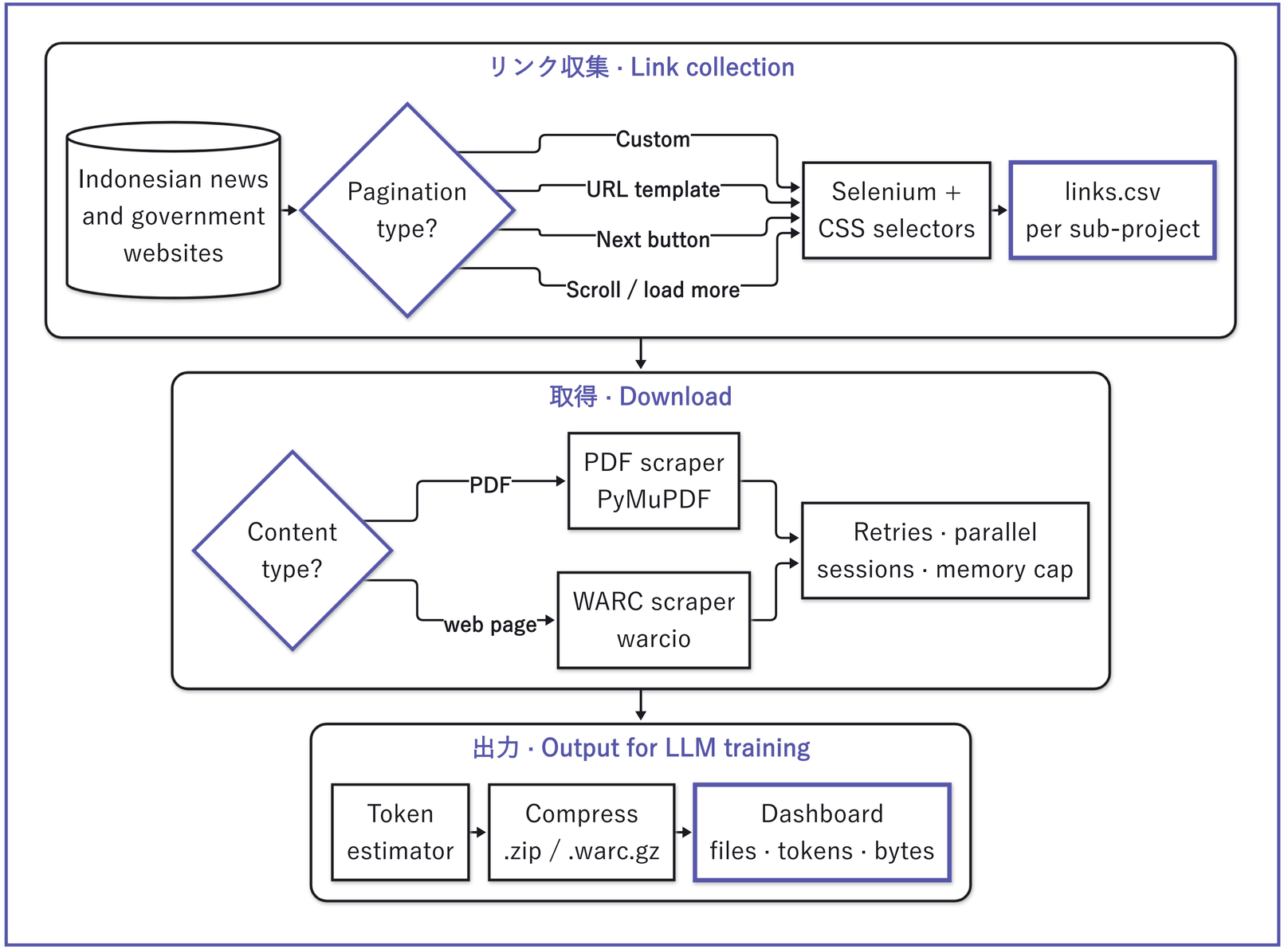
目的	:	Collect Indonesian web pages and PDFs at scale as training data for the Sahabat-AI LLM, handling sites with very different pagination.
言語	:	Python
技術	:	Streamlit, Selenium, BeautifulSoup, PyMuPDF, warcio, Crawl4AI, pandas
メンバー｜期間	:	3 Person｜4 weeks (GoTo Financial, 2025.01)
自分の役割	:	Application architecture; link scraper for all pagination types
アウトプット	:	One app for the whole loop: links, WARC/PDF download, token count, compression and a dashboard.





Left: link scraper with the four pagination strategies. Right: the team at an AI event.

Ez-Scrape：LLM 学習データ収集アプリ



規模 Scale

- The data team gathered 600K+ PDF/WARC files (5B+ tokens) for Sahabat-AI training.

工夫 Design

- Four pagination strategies cover almost every news and government site.
- Each project / sub-project gets its own folders: links, warcs, pdfs, tokens, compressed.
- Retries, parallel sessions and a memory cap keep long scrapes running.

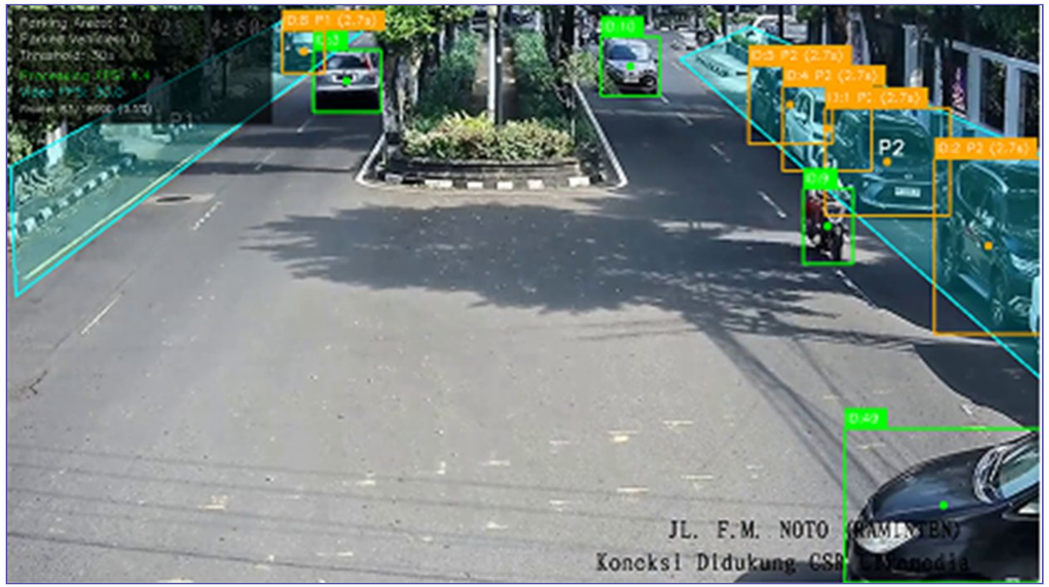
成果 Impact

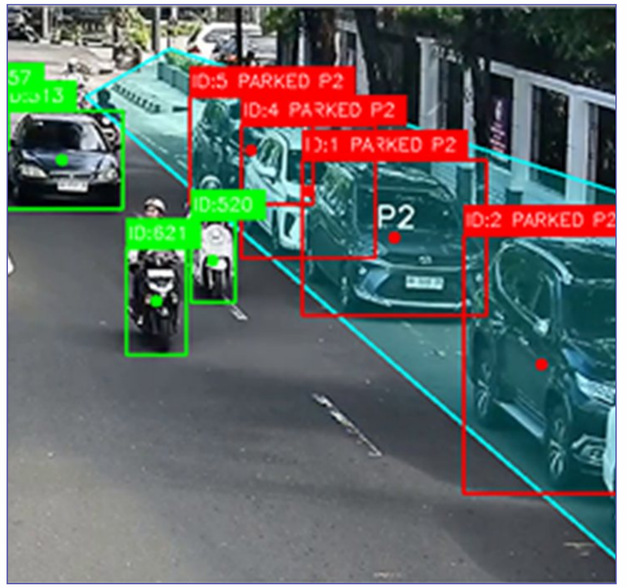
- One reusable app instead of a separate script per website, which made data collection faster.

違法駐車検知システム

GitHub › Nailfaaz/illegal-parking-detection

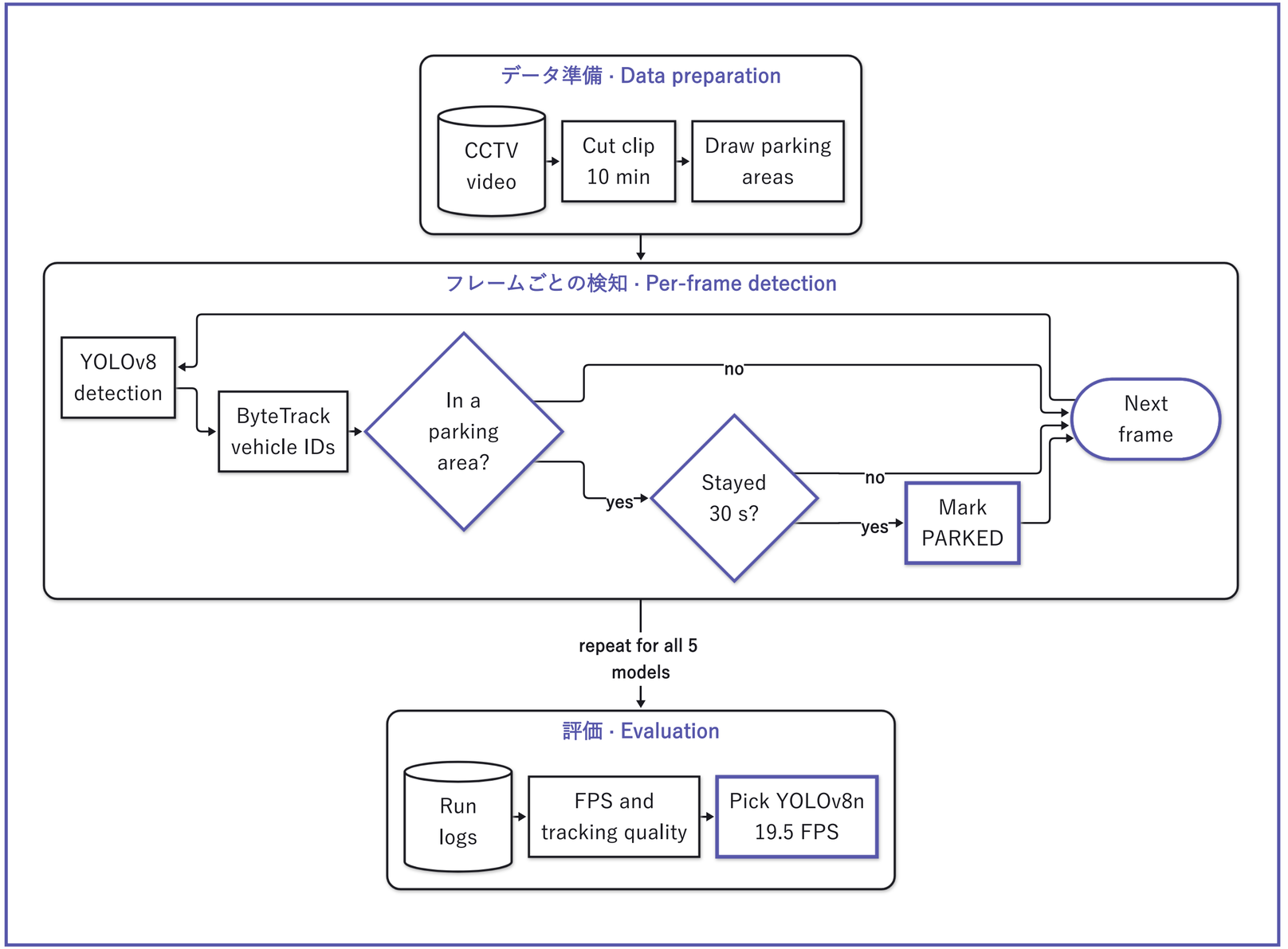
目的	:	Developed an automated illegal parking detection system using YOLOv8, benchmarking multiple models and optimizing performance to achieve real-time inference.
言語	:	Python
技術	:	YOLOv8, OpenCV
メンバー｜期間	:	3 Person｜4 weeks
自分の役割	:	Data Scraping, Model Optimization
アウトプット	:	Vehicle parking detection and tracking system with ~20 FPS real-time inference. 1st place at Datathon 2023 (RISTEK UI), with Best Paper and Best Presentation.







違法駐車検知システム



モデル比較 Benchmark

Model	FPS	Tracking	Overall
YOLOv8n	19.5	95.05	87.49
YOLOv8s	11.6	96.76	80.33
YOLOv8m	6.3	97.96	75.48
YOLOv8l	3.7	98.38	73.06
YOLOv8x	2.8	98.38	72.17

Overall = 0.3 FPS + 0.4 tracking quality + 0.3 detection.

設定 Setup

10-min CCTV clip (18,000 frames at 30 FPS) with 2 parking areas. Confidence $\geq$ 0.5 for cars, motorcycles, buses and trucks.

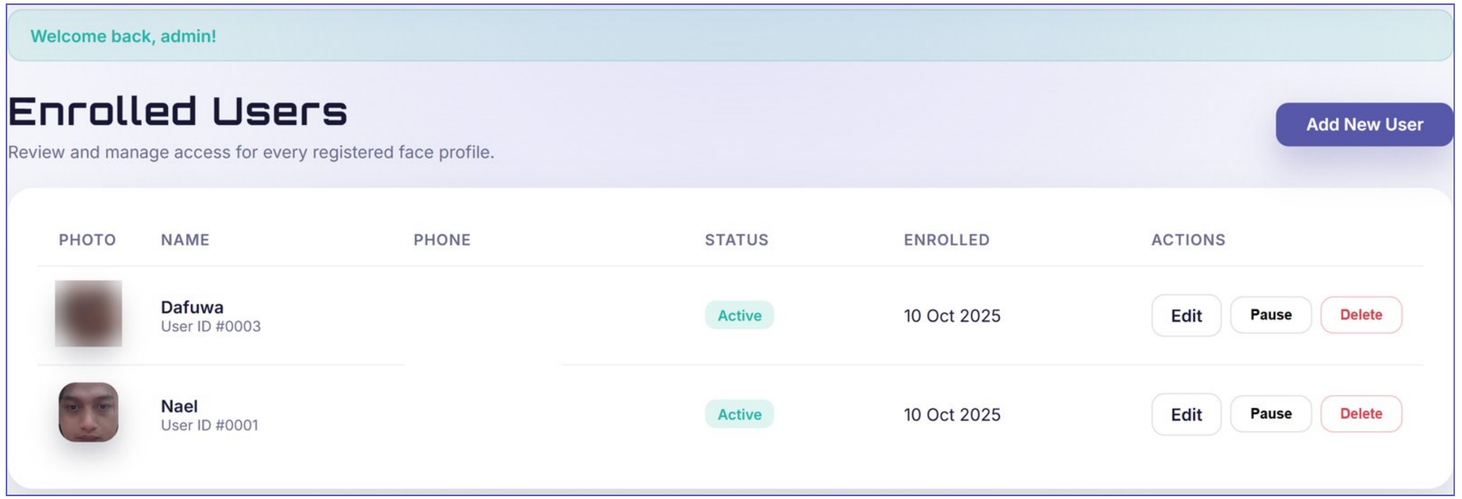
ポイント Findings

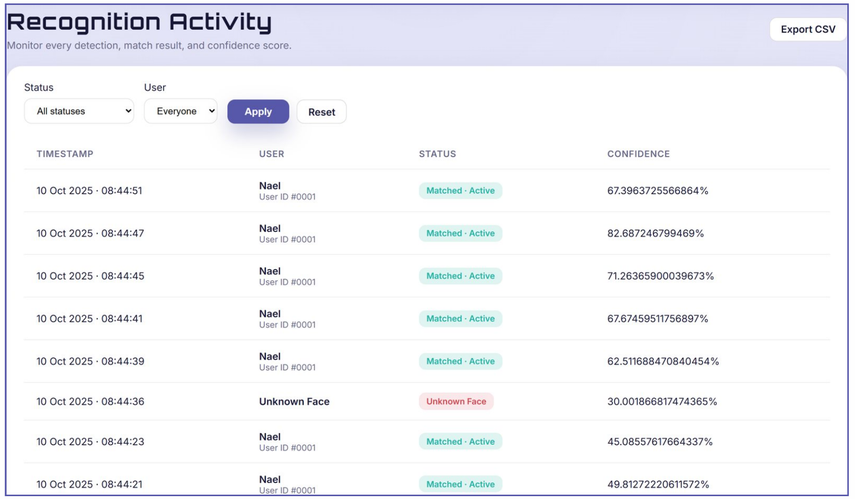
- YOLOv8n: best overall score and the only near-real-time model.
- Larger models track better (95.05 → 98.38) but run ~7× slower.

顔認証入退室管理システム

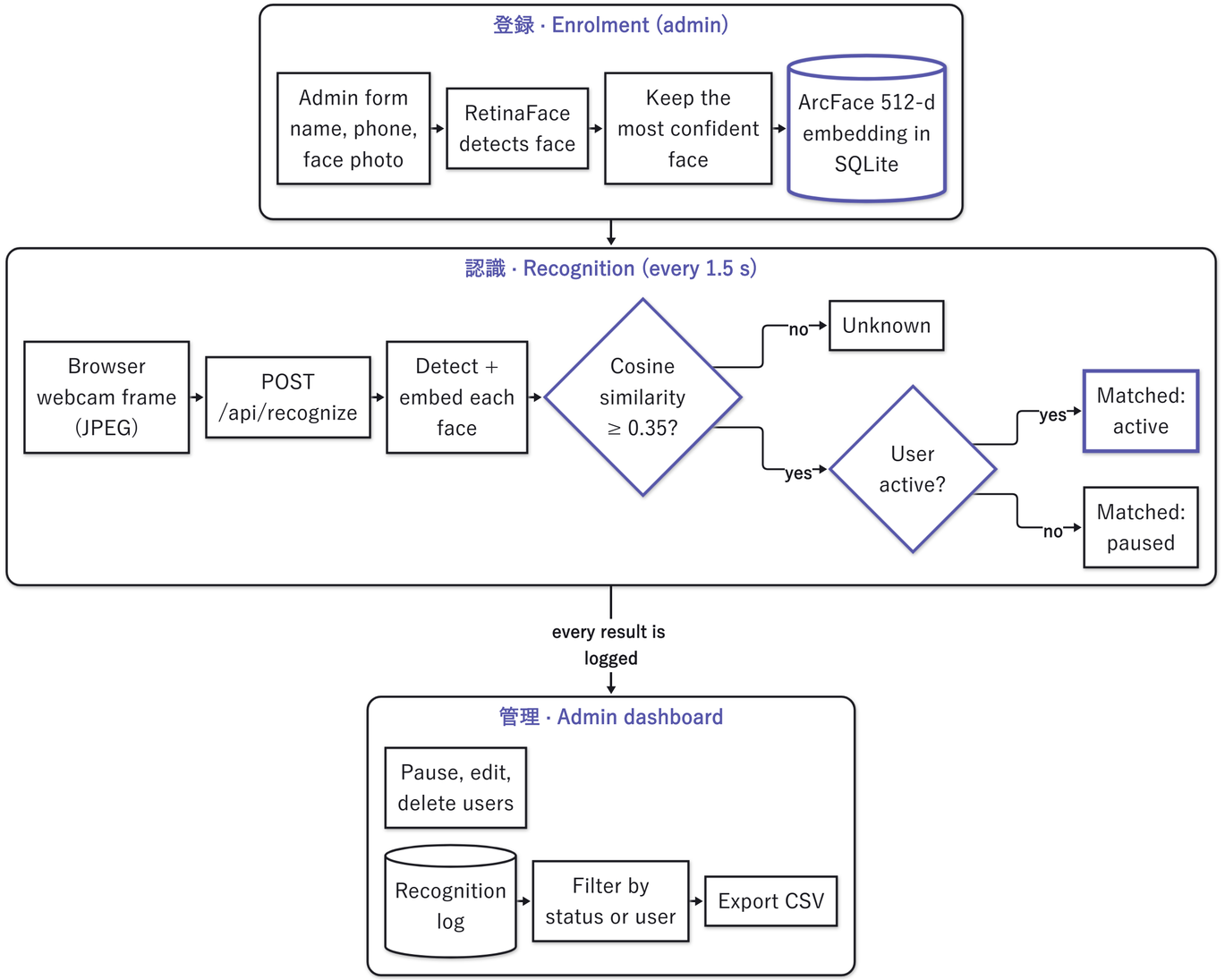
GitHub › Nailfaaz/face-track

目的	:	Build a full-stack face authentication system with secure access control and an admin dashboard for user management.
言語	:	Python, HTML, CSS, JavaScript
技術	:	Flask, SQLAlchemy, SQLite, InsightFace (ONNX Runtime), OpenCV, Flask-Login
メンバー｜期間	:	1 Person｜2 weeks
自分の役割	:	UI/UX design, model tuning, full-stack engineering
アウトプット	:	Web app with real-time webcam recognition, an admin dashboard (enrol, edit, pause, delete) and recognition logs with CSV export.





顔認証入退室管理システム



設定 Settings

- Model: InsightFace buffalo_l (RetinaFace + ArcFace), 640×640 detection, CPU only.
- Match threshold: cosine similarity 0.35.
- Webcam frame every 1.5 s (minimum 0.75 s).
- Each face is stored as 512 float32 values (2 KB).

テスト Test run

Logged outcome	Events
Matched, active	102
Matched, paused	33
Unknown face	10
Total	**145**

安全性 Security

- Admin login with hashed password; HttpOnly session cookie.
- Paused users are still recognised and logged as paused.